



Food web structure and species' role in the sub-Antarctic Marine Protected Area Yaganes

Melina Scian¹ · Luciana Riccialdelli² · Tomás I. Marina²

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Abstract

The Marine Protected Area Yaganes (MPAY) was created in response to the significant deterioration of ecosystems caused by the environmental crisis affecting our planet, with the aim of protecting key ocean habitats and species. Located at the southern tip of South America, MPAY is home to a wide variety of marine biodiversity and habitats, making it of special interest for the conservation of sub-Antarctic environments. For the first time, we described the potential trophic interactions within the MPAY, with a focus on the pelagic realm. We applied both network-level and species-level approaches to characterise the structure and complexity of the food web, aiming to understand trophic dynamics and the specific roles of certain species within it. In terms of complexity, the MPAY food web consists of at least 124 species with 656 predator–prey interactions, a density of interactions of 5.29 and a connectance of 0.04. Regarding its structure, approximately 71% of the species occupy intermediate trophic levels. The Keystone Species Index (KSI) highlighted the importance of some species such as the Patagonian toothfish (*Dissostichus eleginoides*), the amphipod *Themisto gaudichaudii*, euphausiids and myctophids, among others, to the structure of the food web. Identifying the relevance of these species lays the groundwork for the effective stewardship of the little-known MPAY, focussing on its conservation.

Keywords Trophic interactions · Keystone index · Complexity · Structure · Southwest Atlantic Ocean

Introduction

The environmental crisis affecting our planet has caused significant ecosystems deterioration and a dramatic loss of biodiversity and habitats (Sih et al. 2000; Cardinale et al. 2012). In response to these changes, there has been a rapid increase in the number and extent of protected areas globally over the last 20 years (Maestro et al. 2019). This has led to growing interest and action to protect key ocean habitats and species through Marine Protected Areas (MPAs) (Laffoley et al. 2019). MPAs, defined by Dudley (2008) as ‘any area of intertidal or subtidal terrain, together with its overlaying water and associated flora, fauna, historical and cultural features, which has been reserved by law or other effective

means to protect part or all of the enclosed environment’, are a key tool for conserving and restoring marine biodiversity and supporting the sustainable use of marine resources (Pauly et al. 2002; Halpern and Warner 2003; Lubchenco et al. 2003; Tundi Agardy 1994). In this context, the establishment of a representative global network of MPAs is in line with the agreement among nations to protect 30% of marine habitat by 2030 (United Nations 2018).

The conservation of the oceans is a national priority within the framework of the 2030 Agenda for Sustainable Development of the United Nations (United Nations 2018). Argentina is a signatory of the Convention on Biological Diversity (CBD—United Nations 1992), which established the goal of conserving 10% of the Argentine Sea by 2020 (Radovich 2019). In this context, the first Argentine MPA Namuncurá—Burdwood Bank (MPAN-BB I) was created by National Law No. 26,875 in 2013 (Tombesi et al. 2020). Subsequently in 2014, the National System of Marine Protected Areas was established, followed by Law No. 27,490, which was passed in 2018, thus creating two new MPAs: Yaganes and Namuncurá—Burdwood Bank II (MPAY and MPAN-BB II, respectively) (Ferro et al. 2021).

✉ Melina Scian
melinascian@gmail.com

¹ Departamento de Biodiversidad y Biología Experimental, Universidad de Buenos Aires (UBA), Ciudad Autónoma de Buenos Aires, Argentina

² Centro Austral de Investigaciones Científicas (CADIC-CONICET), Ushuaia, Argentina

MPAY is located between the southern tip of South America and the Antarctic Peninsula. Due to its geographical location, the MPAY is influenced by water masses of different origin, such as Cape Horn and Antarctic Circumpolar currents (Guihou et al. 2020). The particular geographic location, combined with its intricate bathymetry (e.g. submarine mountains and canyons), makes MPAY home to a wide variety of marine ecosystems and complex ocean dynamics that foster areas of high productivity and biodiversity. This is key for understanding MPAY as an example of a Large Marine Ecosystem (Guihou et al. 2020).

Ecologically, MPAY is an essential feeding and intense use area for meso and top predators, such as the South American sea lion (*Otaria byronia*) (Milano et al. 2020a) and fur seal (*Arctocephalus australis*) (Milano et al. 2020b), Peale's and dusky dolphins (Dellabianca et al. 2018; Ordoñez et al. 2020), pilot whales (Becker et al. 2021) and baleen and sperm whales (Dellabianca et al. 2023). It is also a common area of globally threatened seabirds, such as the wandering and the grey-headed albatrosses (Falabella 2017). Notably, both albatrosses species, along with the porbeagle shark (*Lamna nasus*), are the emblematic species of MPAY, highlighting its vulnerability due to overexploitation of marine resources (Pauly and Palomares 2005) and habitat degradation (Bax et al. 2003) caused by fisheries. The MPAY is also important for the reproduction of commercially valuable bony fish species, such as the Patagonian toothfish *Dissostichus eleginoides* and the Southern hake *Merluccius australis*) (Troccoli et al. 2020).

The MPAY, together with the MPAN-BB I and II, forms a key component of a protected network in the sub-Antarctic zone. This network is designed to safeguard and enhance the ecological functioning of this southern region, thereby contributing to the overall health of the world's oceans (Riccialdelli et al. 2020; Dellabianca et al. 2023). To achieve this, ecological knowledge on the species and communities in the area is essential, as is understanding the complexity and structure of the network of interspecific interactions. The functional roles of species can be reflected in the interactions they establish (Coux et al. 2016). Therefore, analysing one of the most common relationships between species (Bascompte 2009), as are trophic or predator–prey interactions, based on its empirical properties, provides valuable information about its ecological dynamics (Pascual and Dunne 2006). In particular, topological networks (presence/absence) of trophic interactions have emerged as the most straightforward method for describing the main structural features of complex ecosystems (Thompson et al. 2012). In recent years, network approaches have been successfully applied to study complex high-latitude marine ecosystems, enhancing our understanding of structure, functioning, and response to environmental and anthropogenic changes (e.g. Kortsch et al. 2015; Cordone et al. 2018; Dahood et al. 2020;

Funes et al. 2022; Marina et al. 2024). The MPAY is an incredibly vast, complex and diverse area of international interest due to its geographical location and the physical conditions that characterise it, so establishing a food web with predator–prey interactions with high resolution would be essential to set conservation priorities.

The primary objective of this work was to build and characterise the network of trophic interactions of the Marine Protected Area Yaganes, specifically its pelagic realm, while achieving the highest taxonomic resolution possible. The method used to construct the food web was based on a bibliographic review, so the obtained product is not exhaustive, but represents a baseline and solid knowledge about potential predator–prey interactions and the importance of them for the integrity of the entire ecosystem. To accomplish the objective, we analysed the food web at both (a) the network-level, aiming to provide insights into the complexity and structure of the food web, and at (b) the species-level, to better understand the roles of individual species.

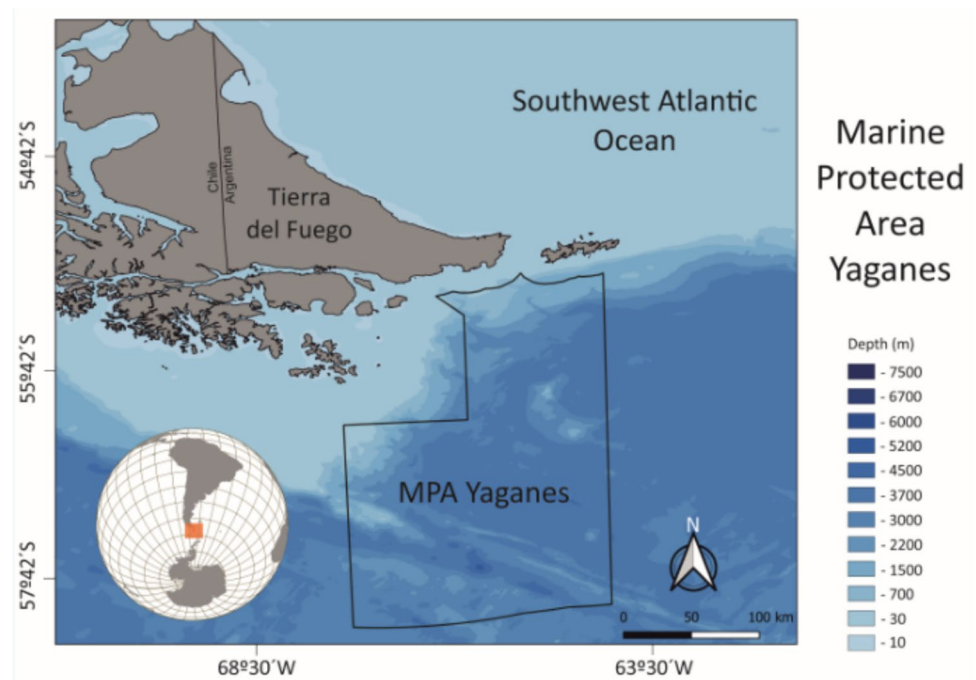
Methodology

Study area

The MPAY is located in the sub-Antarctic region (57.05° S, 65.36° W), south of Tierra del Fuego, in the geographical area of the Drake Passage. The MPAY constitutes the Southwestern end of the Argentine Exclusive Economic Zone, as well as a strategic area of great bio-regional importance as an oceanic corridor between the Atlantic and the Pacific oceans (Tombesi et al. 2020) (Fig. 1). The MPAY covers a marine surface of 68,834.31 km², and spans an area from the continental slope (below the 500 m isobath) to deep oceanic basins (> 3000 m), capturing a diversity of reliefs such as slopes, canyons and seamounts (Palma et al. 2021).

The MPAY receives waters from different origins, creating a complex oceanographic environment. The Antarctic Circumpolar Current (ACC) impacts much of the southern and central sectors. This current flows eastwards and carries cold and nutrient-rich sub-Antarctic waters from the Drake Passage along the upper portions of the western slope of the Argentine basin (Peterson and Whitworth 1989; Piola and Gordon 1989; Acha et al. 2004; Romero et al. 2006). The waters of the ACC mix at its northern boundary with those from the Cape Horn current which also flow Southwest-Northeast (Guihou et al. 2020). Moreover, the northern sector of the MPAY receives waters from the cold estuarial front of the Beagle Channel (Martín et al. 2023).

Fig. 1 Map of the study area, where the Marine Protected Area Yaganes (MPAY) is highlighted (57.05° S 65.36° W)



Construction of the food web

We constructed the network of predator–prey interactions for the MPAY using the species list published in the Argentinean Biodiversity Information System for the area as a starting point (<https://sib.gob.ar/#!/area-prottegida/area-marina-prottegida-yaganes>). Afterwards, we performed a literature search using the name of the species in question and the keywords ‘diet’, ‘prey’, ‘predator’, ‘feeding ecology’ and ‘trophic ecology’ in Google Scholar and Scopus. This process identified more than 50 references including peer-reviewed published articles, Ph.D. theses and public databases (Online Resource 1). Because of the limited information available regarding predator–prey interactions among the species inhabiting and taking advantage of the MPA as a feeding ground, we proceeded with the assumption that if two species interact in an ecosystem similar to that of Yaganes and both coexist in the MPAY, then such interaction was considered valid for the MPAY. Thus, we included predator–prey interactions mentioned by Marina et al. (2024) for the MPAN-BB, and by (López-López et al. 2022) for the Scotia Sea, if both species coexist in the MPAY. Furthermore, in those cases in which two or more taxa presented the same set of predators and prey, the taxa were collapsed into a single trophic species (e.g. Decapoda, Copepoda, Isopoda). In all cases, we considered only adult diets for each species. As a way of validating the list of trophic interactions resulting from the mentioned methodology, we consulted with experts on the different functional groups and species inhabiting the area (refer to Acknowledgements for the experts we contacted). We decided to limit our scope to pelagic, benthic-pelagic

and demersal species due to the lack of information for the benthic habitat, which is linked to the complex bathymetry of the MPA.

Once the compiled list of predator–prey interactions was validated from experts, we transformed this data into a graphical representation, with nodes representing trophic species, edges indicating predator–prey interactions, and arrows pointing by prey to predator. This directional setup mirrors the flow of matter and energy within the ecosystem. Each node represents either a biological species or a group of taxa sharing identical sets of prey and predators, collectively referred to as trophic species (Cohen and Briand 1984).

The network of predator–prey interactions constructed represent potential trophic relationships among pelagic, benthic-pelagic and demersal species without considering spatial and temporal variations. The geographical location and extensive surface, together with the influence of various sea fronts, ocean currents and complex bathymetry make the MPA a highly diverse area, affected by different biological and physical processes (Guihou et al. 2020). Due to this, it is expected that the species diversity, as well as the interactions among them will vary spatially and temporally.

Data analysis

The food web was analysed at two levels: network-level and species-level. The former approach considers all species and interactions within the network, aiming to describe the complexity and structure of the entire food web. The latter analysis considers the interactions and species related to a

particular species, with the objective of describing the role played by the individual species.

Network-level analysis

In order to characterise the food web's complexity and structure, we calculated the following properties: number of species (S), number of interactions (L), density of interactions (L/S), connectance (C), percentage of basal, intermediate and top species, path length (PL), clustering coefficient (CC) and small-world pattern (SW). In a network context, basal species represent the energy sources of the food web (e.g. phytoplankton, detritus), intermediate species present both prey and predators (i.e. primary and secondary consumers), and top species are consumer without predators (Cohen and Briand 1984).

Connectance (C) is defined as the ratio between the number of established interactions (L) and the number of possible interactions in the food web (S^2). This property is a key descriptor of the network structure (Martinez 1992) and a good estimator of the community's sensitivity to disturbances (Dunne et al. 2002a; Montoya et al. 2006).

The path length (PL) is a global network property that measures the distance between species in the food web; here we consider the shortest average distance between species (Watts and Strogatz 1998). The clustering coefficient (CC) is defined as the average fraction of node pairs connected to the same nodes, which are also connected to each other; provides information about the clustering of nodes within their immediate neighbourhood (Delmas et al. 2018). These two network properties are typically analysed in order to gain insight into the small world pattern (SW) (Watts and Strogatz 1998). To determine whether the food web presented this pattern, we compared the empirical values of CC and PL with those resulting from 1000 randomly generated networks created using the Erdős–Rényi model (Erdős and Rényi 1959), with the same number of species (S) and number of interactions (L), following the method proposed by Marina et al. (2018).

Additionally, we studied the degree distribution, which measures the probability that a species has k interactions within the network. This is calculated as $P(k) = N(k)/S$, with $N(k)$ being the number of nodes with k interactions (Albert and Barabási 2002). We calculated the cumulative degree distribution for the food web, and fitted it to exponential, power law and truncated power law distribution models, using the AIC values as decision criterion.

From the analysis of this property, important nodes of the network can be identified, such as potential generalist species (i.e. species with a diverse diet, that is, with a high number of interactions), and specialists (i.e. species with a narrow diet, that is, with a low number of interactions). We

considered both in-degree (i.e. number of preys) and out-degree (i.e. number of predators) of each species.

Species-level analysis

The species-level approach focussed on determining the individual role of species in the context of the food web, given that not all species in large communities fulfil the same ecological role, nor are they equally important for network processes and properties (Delmas et al. 2018). For this, we considered the following properties: (a) degree, (b) closeness, (c) betweenness, and (d) trophic level. The first three correspond to centrality properties, that is referring to a particular node in the network and its relationship with the rest of the nodes, and quantify the influence of a species within the network (Delmas et al. 2018). The degree is the total number of interactions of a species, considering in- (number of predators) and out-interactions (number of prey) (Jordán 2009); it quantifies the immediate influence between species (Delmas et al. 2018). Those species with a high degree are the most connected, so their influence on the robustness (i.e. fraction of primary species loss that induces at least 50% total species, Cordone et al. 2018) of the network is high. The extent of trophic connections a species holds within a food web, the greater its potential to affect community structure (Dunne et al. 2002b).

Closeness measures the proximity of a species to all other species in the network. It is based on the shortest path between two species and therefore indicates how efficiently perturbations about a species are likely to influence the network (Delmas et al. 2018). Regarding betweenness, it is defined as the number of times that a species is found between a pair of other species in the network (Delmas et al. 2018), being an ideal measure to study the influence of species loss in fragmentation processes (Earn et al. 2000; Chadès et al. 2011; McDonald-Madden et al. 2016), or in the face of the dispersion of disturbances.

The trophic level measures the distance, in number of interactions, between a species and the basal species of the network (Lindeman 1942). It represents the number of feeding links separating an organism from the base of production (Thompson et al. 2007), and indicates basal, intermediate and top species.

Centrality indices have been used to identify possible key species in ecological networks (Jordán and Scheuring 2004; Martín González et al. 2010); key species being understood as those that interact strongly (Davic 2003) and are disproportionately important in their community (Mouquet et al. 2013). Based on this, we created a Keystone Species Index (KSI). To do this, we ranked species according to each centrality property: the higher the degree, closeness and betweenness values, the higher the species rank. Then we averaged these three rankings to obtain the KSI for every

species. In this way, the KSI takes values from 1 (species with the highest centrality indices, first position of each ranking) to the number of species in the network (species with the lowest centrality indices, last position in each ranking). When two or more trophic species have the same KSI value, they share the same position in the ranking, which means a top-10 ranking could include more than 10 species. Furthermore, we explored the relationship between the KSI and the trophic level, with the aim of gaining insights into the trophic position of the relatively most important species in the network. Thus, we considered the trophic level as the dependent variable and the KSI as the independent variable. Fitting was done locally by least-squares (family = gaussian), meaning that for the fit at point x , the fit is made using points in a neighbourhood of x , weighted by their distance from x . The size of the neighbourhood is controlled by α ; here equal to 0.75. For $\alpha < 1$, the neighbourhood includes the proportion α of the points, and these have tricubic weighting (proportional to $1 - (\text{dist}/\text{maxdist})^3$) (Cleveland et al. 1992).

Analyses were carried out using R software (version 4.1.2, R Core Team 2023), particularly the packages igraph version 1.4.2 (Csárdi et al. 2023), multiweb version 0.6.8

Table 1 Network-level properties characterising the structure of the MPAY food web

Basal	Intermediate	Top	PL	CC	SW
4.03%	70.97%	25.00%	3.11	0.21	Yes

Basal, intermediate, top: percentages of basal, intermediate and top species

PL path length, CC clustering coefficient, SW small world pattern

(Saravia 2023) and NetIndices version 1.4.4.1 (Kones et al. 2009). The source code and data are available at the following GitHub repository: <https://github.com/melinascian/AMP-YAGANES.git>.

Results

Network-level analysis

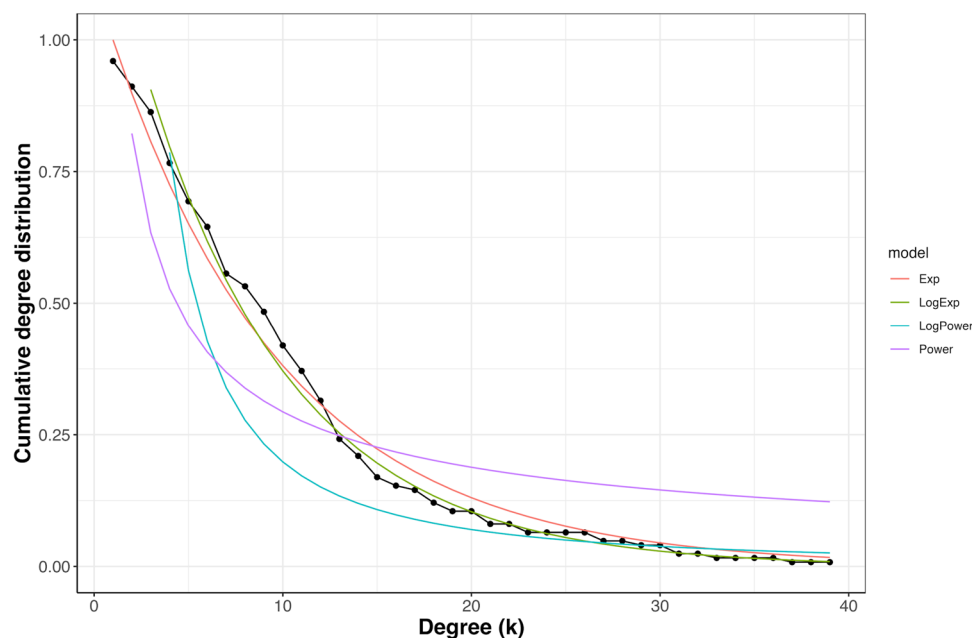
The MPAY food web consisted of at least 124 species identified from predator–prey interactions system, species with 656 documented predator–prey interactions, an interaction density of 5.29, and a connectance of 0.04. Almost two-thirds of the species occupied intermediate trophic levels (70.97%), and the majority of the species in the food web were consumers, 119 out of 124. The food web displayed a small-world pattern (Table 1), with a shorter path length and a higher clustering coefficient than random networks (Online Resource 2).

The degree distribution of the food web showed an asymmetric frequency in the number of interactions per species, which means that most species had few interactions while a few species concentrated the majority of them (Fig. 2). The cumulative degree distribution of the marine food web fitted the best to an exponential model (Online Resource 3).

Species-level analysis

The top-five species with the highest degree (number of total interactions considering both predators and prey) were: the Patagonian toothfish (*Dissostichus eleginoides*, 42

Fig. 2 Cumulative degree distribution for the MPAY food web. Degree (k) is the probability of species to present k number of interactions. Best fit is the exponential model (refer to Table S3 for model statistics)



interactions), *Euphausiidae* (37 interactions), the Patagonian squid (*Doryteuthis gahi*, 33 interactions), the Fuegian sprat (*Sprattus fuegensis*, 31 interactions) and amphipod (*Themisto gaudichaudii*, 31 interactions), and the Southern blue whiting (*Micromesistius australis*, 29 interactions) along with the Southern longtail cod (*Patagonotothen ramsayi*, 29 interactions). Considering the in-degree, the top-five predators in number of preys were: Patagonian toothfish (29 preys), false killer whale (*Pseudorca crassidens*, 23 preys), Peale's dolphin (*Lagenorhynchus australis*, 19 preys), and Southern long-finned pilot whale (*Globicephala melas*, 19 preys), fin whale (*Balaenoptera physalus*, 18 preys) and Southern longtail cod (17 preys), along with one of MPA emblematic species the porbeagle shark (*Lamna nasus*, 17 preys). On the other hand, the top-five prey in number of predators (higher out-degree) were: *Euphausiidae* (35 predators), the Patagonian squid (29 predators) and the amphipod *T. gaudichaudii* (29 predators), the Fuegian sprat (25 predators), the squat lobster (*Grimothea gregaria*, 24 predators), and the Southern blue whiting (18 predators) (Online Resource 2).

The top-five species with the highest values of closeness were: the Patagonian toothfish (0.00446), the amphipod *T. gaudichaudii* (0.00427), *Euphausiidae* (0.00424), the squat lobster (0.00422), and the Patagonian squid (0.00417). In regard to betweenness, species with the highest values were those who participated in the highest number of shortest paths between species; the 5 species were: Patagonian toothfish (1521), bush-club squid (*Batoteuthis skolops*, 1034), cock-eyed squid (*Histioteuthis eltaninae*, 858), whip-lash squid species (*Mastigoteuthis* spp, 700), and the Southern blue whiting (444) (Fig. 3) (Online Resource 2).

The KSI ranking shows the top 10 most relevant positions for the MPAY food web, with several species sharing the same rank (e.g. *Patagonotothen ramsayi* and *Micromesistius australis* share position 2, Table 2). An unimodal pattern is observed in the relationship between KSI and trophic level (TL), where KSI values are generally high between TL 2.5 and around TL 5.5 (Fig. 4). There were several species of mid-trophic levels, such as fishes and cephalopods, with high KSI values, as well as top predators like toothfish and Peale's dolphin. *Euphausiids* and squat lobsters also exhibited high KSI values.

Discussion

The structure of the communities inhabiting the MPAY is a consequence of a complex and diverse oceanographic environment influenced by water of different origins (Voigt et al. 2013; Guihou et al 2020). The MPAY food web characteristics are aligned with high-latitude marine ecosystems ones (e.g. Potter Cove), suggesting a wasp-waist control and a

small world pattern (e.g. Riccialdelli et al. 2020; Büring et al. 2024; Marina et al. 2024). Also, we identified certain species considered as top predators, such as *Dissostichus eleginoides*, as the most influential species in the structure and functioning of the MPAY.

Food web complexity and structure

The network-level approach, particularly through various metrics of food web topology, provided us with insights into their vulnerability to perturbations (Pimm 1982; May 1986; Link 2002; Cohen et al. 2012), aiding in understanding the flows of energy and matter among organisms (Rodriguez et al. 2022). Connectance is a key descriptor of network structure (Martinez 1992) and a good estimator of the community's sensitivity to perturbations (Dunne et al. 2002a; Montoya et al. 2006). MPAY showed a low connectance, consistent with food webs from high-latitude regions (e.g. Potter Cove, Beagle Channel, Burdwood Bank) which tends to exhibit a connectance closer to the minimum (between 0.01 and 0.05) (Santana et al. 2013; Kortsch et al. 2019; Rodriguez et al. 2022). Low connectance networks (<0.05) display extreme sensitivity to loss of the most connected nodes, but this sensitivity decreases as the connectance increases (Estrada 2007). A low connectance would often be associated with reduced stability (López-López et al. 2022), particularly lower robustness to secondary extinctions or resistance to invasions (Dunne et al. 2002a). According to Dunne et al. (2002b), the removal of less than 25% of species produces the complete collapse of the lowest connectance webs.

There exists a relationship between the shape of the degree distribution of a food web and its connectance (Dunne et al. 2002a). The MPAY food web, besides exhibiting low connectance, displays an exponential degree distribution, indicating an asymmetric frequency of interactions with few highly connected species, and most species being poorly connected. The degree distribution can determine the system vulnerability to the loss of the most connected species (Albert et al. 2000; Estrada 2007): those food webs with a right-skewed distribution, as observed in the case of MPAY, are more susceptible to the removal of highly connected species (Dunne et al. 2002a; Estrada 2007). Specifically, in the MPAY, these species tend to occupy upper and mid-trophic levels such as *Dissostichus eleginoides*, *Doryteuthis gahi* and *Sprattus fuegensis*, within others. In particular, disturbances affecting generalist species lead to secondary extinctions, while when a specialist is disturbed, the network is able to withstand a larger disturbance before an extinction (McCann and Hastings 1997; Rodriguez et al. 2022).

Understanding the mechanisms that shape the trophic structure is crucial. Depending on the dominant control type

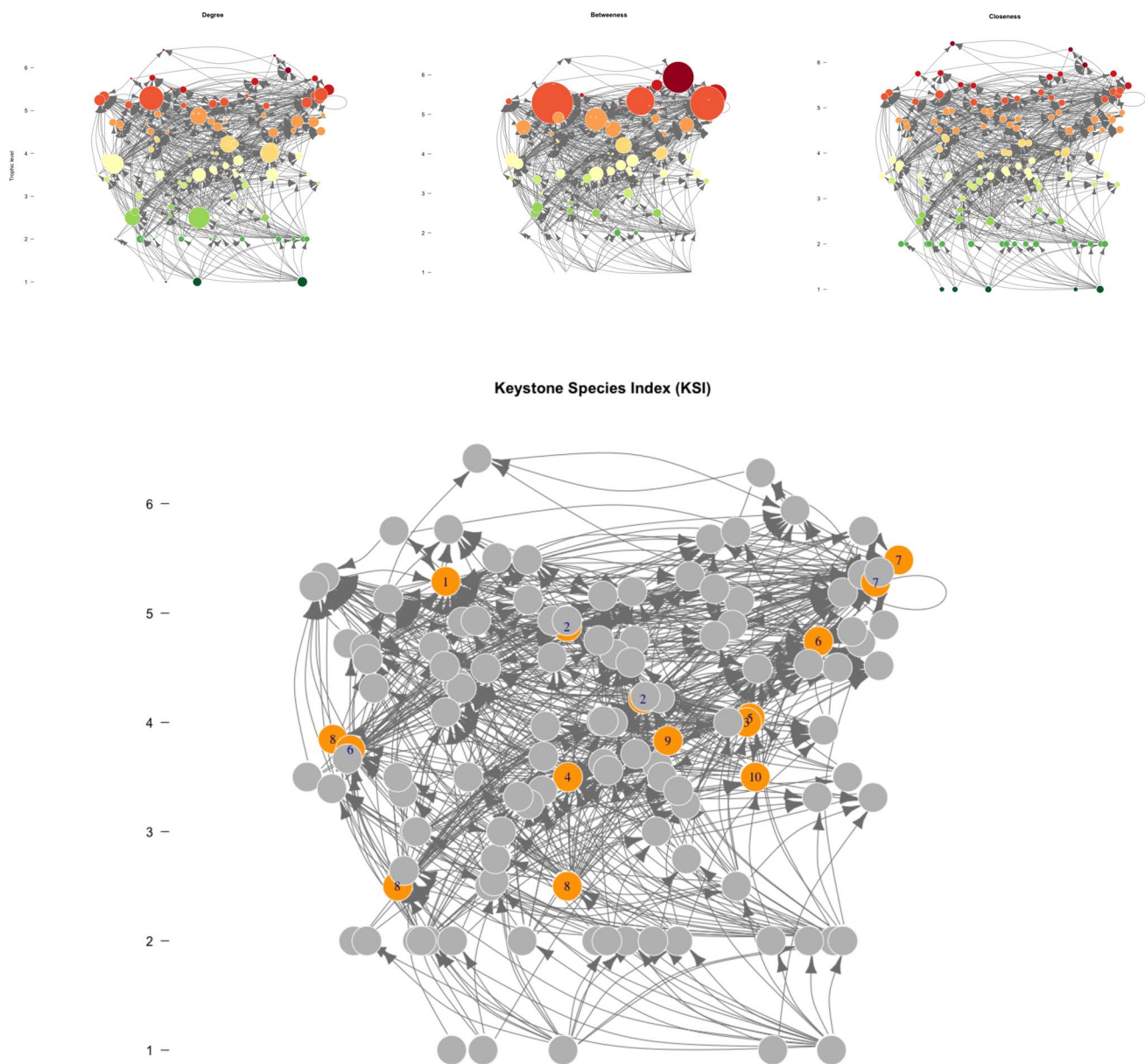


Fig. 3 Graphic representation of the relative importance of the species of the MPAY food web, according to the three centrality indices (degree, betweenness and closeness) and the Keystone Species Index (KSI). In the index graphs the diameter of the circles (species) indicates the relative importance of the species, the colour gradient refers

to the species' trophic level (dark green=basal species, dark red=top species). In the graph showing the KSI, the most important species are highlighted in orange colour and their ranking is shown (refer to Table 2)

(e.g., bottom-up, top-down or wasp-waist) the propagation of disturbances throughout the food web might vary, thus affecting how populations respond to perturbations (Cury 2000; Hunt and McKinnell 2006; Young et al. 2015). The food web structure is a dynamic process intricately linked to the presence, the abundance and distribution of species in a community (Young et al. 2015); the prevailing type of control significantly influences this dynamic process. 'Wasp-waist' (WW) controlled systems are primarily regulated by a small number of mid-trophic level species (Ricciardelli

et al. 2020; Büring et al. 2024). These species present a much higher abundance than the rest of the species of the ecosystem, playing a crucial role in shaping the structure of the food web by influencing both their potential predators and their own prey (Cury 2000; Bakun 2006). Our results indicate that several of the most important species (i.e. top 10 of the Keystone Species Index), present mid-trophic level positions in the food web (TL = 3–4.5). Moreover, some of these species such as the Southern longtail cod, amphipod *Themisto gaudichaudii* and the Fuegian sprat present high

Table 2 Ranking of the most important species (top 10) according to the Keystone Species Index (KSI), obtained from the centrality indices: degree, closeness and betweenness

Trophic species	KSI	Ranking	TL
<i>Dissostichus eleginoides</i>	1.00	1	5.29
<i>Patagonotothen ramsayi</i>	6.33	2	4.22
<i>Micromesistius australis</i>	6.33	2	4.88
<i>Themisto gaudichaudii</i>	7.00	3	4.00
<i>Gymnoscopelus spp.</i>	7.67	4	3.50
<i>Sprattus fuegensis</i>	8.33	5	4.04
<i>Doryteuthis gahi</i>	9.00	6	3.75
<i>Lagenorhynchus australis</i>	9.00	6	4.74
<i>Salilota australis</i>	10.33	7	5.48
<i>Batoteuthis skolops</i>	10.33	7	5.28
<i>Patagonotothen guntheri</i>	11.00	8	3.85
<i>Euphausiidae</i>	11.00	8	2.50
<i>Grimothea gregaria</i>	11.00	8	2.50
<i>Enteroctopus megalocyathus</i>	15.67	9	3.83
<i>Electrona spp.</i>	16	10	3.50

The trophic level (TL) of each species is provided

densities in the sub-Antarctic region (Diez et al. 2016, 2018; Fischer et al. 2022). Overall, this aligns with the WW control model proposed by Riccialdelli et al. (2020) and Büring et al. (2024) in neighbouring marine areas. However, trophic regulation can be much more complex, with various forces acting simultaneously (Hunt and McKinnell 2006). Thus, the influence of top predators, such as *Dissostichus eleginoides*, represent a top-down force that may also be significantly affecting the MPAY.

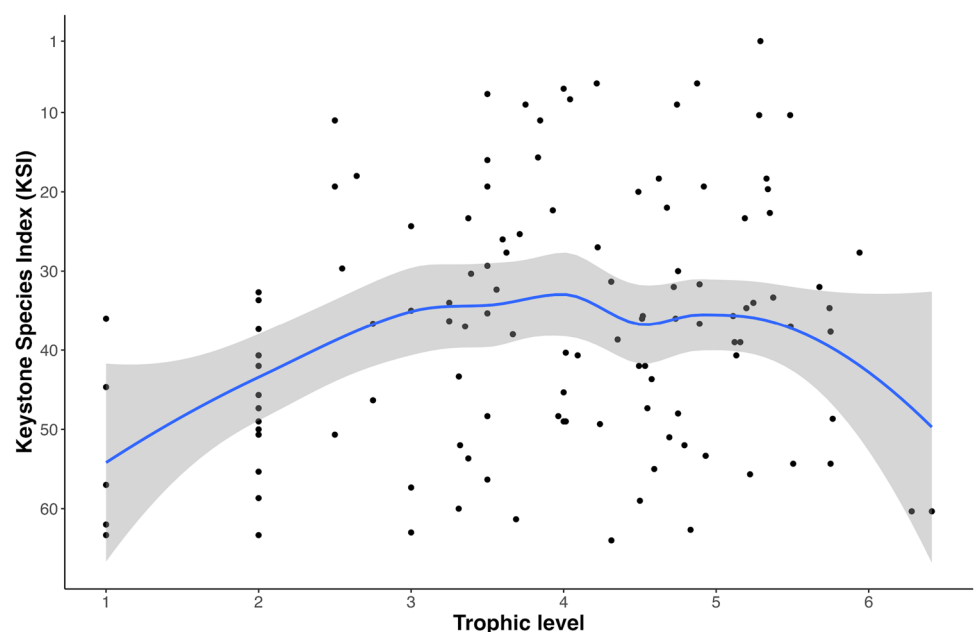
The small-world topology, a pattern that gives a clear overview of organisation and resistance in ecological networks (Bornatowski et al. 2017), is characterised by lower path length (PL) and higher clustering coefficient (CC) values than those obtained for random webs. PL and CC values obtained for MPAY food web influence the behaviour of it: a short PL implies a rapid propagation of an impact (e.g. biodiversity loss, marine contamination, overfishing) throughout the food web (Delmas et al. 2018), while high CC allows a rapid response to disturbances (Montoya and Solé 2002), which influence the resilience of the communities against perturbations (Gilarranz et al. 2017; Stouffer and Bascompte 2011). It has been suggested that the robustness of food webs displaying a small-world pattern might be lower in face to the loss of highly connected species (e.g. Gaichas and Francis 2008; Navia et al. 2016; Bornatowski et al. 2017), regarding its susceptibility.

Exploring the role of consumers in the food web

The species-level approach highlights that most of the predators in the MPAY food web are specialists and only a few consumers have many prey. The most evident generalists are the Patagonian toothfish (KSI = 1) and the false killer whale (KSI = 33), with more than 20 potential prey. Both species act as top predators (TL = 5.29 and 5.37, respectively).

The significance of the primary consumers in the food web is highlighted by the presence of the top-five prey species in terms of predator number, all within the top-10 of the KSI ranking. This functional group serves as a keystone link between primary producers and higher consumers (Zhu and Zhu 2022). Among these species, the most demanded prey

Fig. 4 Relationship between the Keystone Species Index (KSI) and the species' trophic level. The blue line and grey shaded area represent the average and confidence interval 95%, respectively, for the locally estimated scatterplot smoothing (loess) local fitting we performed. Note that the y axis (KSI ranking) is inverted. Summary of the fitting: number of observations (124), equivalent number of parameters (4.64), residual standard error (1.288), neighbourhood size (0.75), degree of polynomials (2), family: gaussian



in the MPAY were the euphausiids (KSI = 10.66), which are key components in productive regions of temperate and high-latitudes ecosystems (Sabatini et al. 2012; Presta et al. 2023). The availability of these primary consumers species has profound consequences for consumers that depend on them (Ricciardelli et al. 2020), as they constitute a crucial energy channel from lower to higher trophic levels.

Keystone species and their role in the community

Ecosystems are characterised by presenting food webs in which the position of certain species is of relevance for the entire community, which allows us to identify them as key species (Jordán 2009). These key species, due to their trophic role, influence how the community responds to disturbances (e.g. climate change), affecting its resilience, robustness and vulnerability. Therefore, their identification is an essential first step for proper management of the community towards its conservation. The species present in the top-10 KSI ranking turned out to have a fundamental role in the pelagic community of the MPA, either by being the most connected (high degree), frequently found between two other species in the food web (high betweenness), or being closest to all other species (high closeness). This ranking is predominantly occupied by fishes (*Dissostichus eleginoides*, *Patagonotothen ramsayi*, *Gymnoscopelus* spp., *Sprattus fuegensis*, *Micromesistius australis*, *Salilota australis*, *Patagonotothen guntheri* and *Electrona* spp.), followed by crustaceans (*Themisto gaudichaudii*, *Euphausiidae* and *Grimothea gregaria*), cephalopods (*Doryteuthis gahi*, *Batoteuthis skolops* and *Enteroteuthis megalocyathus*) and a single species of cetacean (*Lagenorhynchus australis*).

The Patagonian toothfish turned out to be the first in the KSI ranking. It is a demersal fish (Collins et al. 2010), characterised by moderate growth rates (Troccoli et al. 2020), long lifespan (Yates et al. 2018) and high vulnerability to overexploitation (Gilarranz et al. 2017). Its life cycle includes different ontogenetic phases: adults are typically found in deeper waters, while larvae and post-larvae presence is registered in pelagic layers (North 2002), leading to different feeding patterns of each life stage (Arkhipkin et al. 2003). Here, only the adult stages were considered, as the presence of earlier stages is rare in the deep waters of the MPAY. This fish presents the highest value of degree and betweenness, meaning not only that it is the most connected species, but also the one that acts as “bridge” for energy transfer, given that it participates in relatively more short food chains, affecting a greater number of energy flows (Lai et al. 2012). Furthermore, it is the species with the highest closeness, indicating that any disturbance that affects this species will propagate through the entire food web more rapidly, disrupting its functioning and stability (Estrada and Bodin 2008). Fisheries in the Argentine Exclusive Economic

Zone (EEZ) began in the early 1990s, and quickly experienced significant growth due to the high commercial value of the species on international markets (Martínez et al. 2023), with the development of a Patagonian toothfish fishery south of 47° S (Céspedes et al. 2016). Currently, the resource is exploited throughout its circumpolar Antarctic distribution, including both Antarctic waters and areas surrounding the peninsular platforms of sub-Antarctic islands, such as Malvinas or South Georgia Islands (Galleguillos et al. 2006). The overexploitation of this species, particularly due to improper management, could pose a threat to the species (Collins et al. 2010), affecting the integrity of the MPAY pelagic community, given the importance of the notothenioid on it.

The Southern longtail cod ranked second in KSI. It plays a key role in the South-west Atlantic shelf food web both acting as prey and predator (Brickle et al. 2006; La Mesa et al. 2016; Ricciardelli et al. 2020). In particular, this species constitutes a significant component as a WW species in a nearby area, the MPA Namuncurá-Burdwood Bank, owing to its high abundance, trophic position, and significance as food source for predators (Arkhipkin and Laptikhovsky 2013; Ricciardelli et al. 2020). Our results suggest that a similar ecological control exerted by the species in the MPAY is expected.

The Antarctic myctophids of the genus *Gymnoscopelus* and *Electrona* were positioned high in the overall KSI ranking. Although these mesopelagic fishes, which are dominant in terms of diversity and abundance in most of the world's oceans (Gjøsaeter and Kawaguchi 1980), play a crucial role in pelagic food webs, their specific function in this context has been poorly studied (Saunders et al. 2015, 2019; Van der Grient et al. 2023). They are thought to eat mostly primary consumers (Shreeve et al. 2009), and be a major source for higher predators (Connan et al. 2007), existing a major flux of secondary productivity passing through them (Saunders et al. 2019). Moreover, myctophids have been found to be important as an alternative trophic pathway in Southern Ocean food webs centred upon Antarctic krill in seasons of low krill abundance (Murphy et al. 2006) (e.g. Scotia Sea), where the structure of the food web varies seasonally in response to changes in physical and chemical processes that modified primary production and zooplankton dynamics (Saba et al. 2014). Considering the MPAY an austral and oceanic ecosystem, it is expected that myctophids have a similar role in the area, contributing to overall ecosystem stability and resilience (Saunders et al. 2019). This is particularly important in the context of current climate change, as a decline in krill populations in Antarctic and sub-Antarctic regions has been suggested (Flores et al. 2012), along with an increased fishing pressure (Watters et al. 2020). As a result, the role of myctophids in maintaining food web stability may increase under this scenario (Saunders et al. 2019).

Other fish species ranked in the top-10 KSI are considered important ecological components, valuable commercial resources, or both. This is the case of the Southern blue whiting, a pelagic-demersal fish (Wöhler et al. 2001) which is the target species of Argentinian surimi vessels, primarily fishing east of Tierra del Fuego; and the Fuegian sprat, which is frequently referred to as the largest potential pelagic resource on the austral extreme of the sea shelf off Argentina (Sánchez et al. 1995). Furthermore, the Patagonian cod and the yellow notothenioid, both minor fishing resources in the Southwest Atlantic Ocean (Gorini et al. 2007; Collins et al. 2008), are also species with a relevant role in the MPAY food web.

Several crustacean species have been previously identified as WW species for the southwestern South Atlantic Ocean (Ricciardelli et al. 2020). In line with this, the hyperiid amphipod *T. gaudichaudii*, the squat lobster and euphausiids ranked in the top-10 KSI. The former one is a major component of the macrozooplankton community of the Argentinian shelf off Southern Patagonia (Sabatini and Alvarez Colombo 2007), and is considered a key species in sub-Antarctic food webs (Padovani et al. 2012). Particularly, it is a main food source for the austral fish assemblage (south of 47°) (Collins et al. 2008; Padovani et al. 2012). Noteworthy, the consumption of *T. gaudichaudii* varies spatially, which has been suggested to be related to an increase of primary production caused by the presence of oceanic fronts and their associated circulation patterns (Palma et al. 2004; Sabatini et al. 2004). The squat lobster is a benthic crustacean species that also forms pelagic swarms (Diez et al. 2016). It is considered an ecologically important species due to its role in energy transfer to higher trophic levels (Lovrich and Thiel 2011; Ricciardelli et al. 2020; Rodriguez et al. 2022; Marina et al. 2024), and a potential fishery resource in sub-Antarctic waters of South America (Diez et al. 2018). An expansion of this decapod along the Argentine Continental Shelf and Beagle Channel has been documented over the past 15 years (Diez et al. 2016), likely linked to variations in the productivity of frontal zones in the area, possibly associated with climate change. Finally, krill (e.g. *Euphausia superba*, *Euphausia vallentini*) represents a main food source for most high-trophic level predators in Antarctic and sub-Antarctic systems (Collins et al. 2010).

Regarding to cephalopods, the Patagonian squid, the only species of *Doryteuthis* found in the Southern Atlantic Ocean (Büiring 2023), and one of the main components of marine food webs linking primary consumers and marine top predators (Gasalla et al. 2010; Coll et al. 2013; Navarro et al. 2013), ranked sixth at KSI ranking. This species has recently been identified as an abundant mid-trophic species that plays an important role in the Malvinas/Falkland shelf ecosystem that also exhibits a WW control (Büiring et al. 2024).

The top-10 KSI ranking species composition suggests that the MPAY is a WW ecosystem, given that more than half of the species present a trophic level within the range 3–4.5. This is consistent with the ‘wasp-waist’ control hypothesis described by Ricciardelli et al. (2020) and proposed to dominate at the southwestern South Atlantic Ocean (e.g. Padovani et al. 2012; Arkhipkin and Laptikhovsky 2013; Saporiti et al. 2015; Ricciardelli et al. 2020; Van der Grient et al. 2023; Büiring et al. 2024; Rodriguez et al. 2024). Under this control, a few mid-trophic level species link upper and lower trophic levels, channeling the energy flow through the food web (Cury 2000; Bakun 2006). An aspect to consider is that, due to the physical characteristics of the MPAY (bathymetry, currents and sea fronts that influence it, etc.), it is expected that, although the entire area may be under a WW control, the dominant species (i.e. top-10 KSI species) will vary.

Since this work is based on a literature review, and the study area (MPAY) is relatively understudied, it is almost certain that some species have been overlooked. This may affect the comprehensiveness of the analysis and lead to an underrepresentation of certain groups, particularly those for which there are data gaps (e.g. benthic community). While the focus of this study was primarily on the pelagic realm, incorporating data from other realms would further enrich the analysis presented here. For instance, it would be crucial to study the benthic community across all the MPA, and the way it interacts with the pelagic habitat, to better understand the ecological dynamics of such a complex ecosystem. Nevertheless, the information we provided is crucial as a foundation for the effective management of MPAY.

Conclusion

Our study represents the first step in elucidating the network of trophic interactions within the Marine Protected Area Yaganes, which is crucial for managing such a commercially and ecologically important area. The combined analysis, at both the network and species levels, provided a detailed insight into the complexity and structure of this ecosystem. Our results suggest that the structure of pelagic ecosystems is more complex than previously thought (López-López et al. 2022). We suggest that the MPAY food web is primarily under a wasp-waist control. Our analysis at the species-level highlights not only the role of mid-trophic level species, such as the amphipod *T. gaudichaudii*, myctophids, the Fuegian sprat, and the Patagonian squid, in maintaining the integrity of the food web but also the role of Patagonian toothfish, considered a top predator. Describing species' roles in food webs provides a toolbox with which to assess the significance of species in terms of community's functioning and stability (Thébault and Fontaine 2010; Cirtwill et al. 2018; López-López et al. 2022). Then, conserving

important species, as the ones identified in this study for the MPAY, and mitigating negative effects of climate change on these, will be crucial for managing diversity loss and ensuring its conservation.

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Data availability The source code and data are available at the following GitHub repository: <https://github.com/melinascian/AMP-YAGAN-ES.git>.

Declarations

Conflict of interest The authors declare no conflicts of interest.

References

- Acha EM, Mianzan HW, Guerrero RA, Favero M, Bava J (2004) Marine fronts at the continental shelves of austral South America. *J Mar Syst* 44:83–105. <https://doi.org/10.1016/j.jmarsys.2003.09.005>
- Albert R, Barabási AL (2002) Statistical mechanics of complex networks. *Rev Mod Phys* 74(1):47–97. <https://doi.org/10.1103/RevModPhys.74.47>
- Albert R, Jeong H, Barabási AL (2000) Error and attack tolerance of complex networks. *Nature* 406:378–382. <https://doi.org/10.1038/35019019>
- Arkhipkin A, Brickle P, Laptikhovsky V (2003) Variation in the diet of the Patagonian toothfish with size depth and season around the Falkland Islands. *J Fish Biol* 63(2):428–441. <https://doi.org/10.1046/j.1095-8649.2003.00164.x>
- Arkhipkin A, Laptikhovsky V (2013) From gelatinous to muscle food chain: rock cod *Patagonotothen ramsayi* recycles coelenterate and tunicate resources on the Patagonian Shelf. *J Fish Biol* 83:1210–1220. <https://doi.org/10.1111/jfb.12217>
- Bakun A (2006) Wasp-waist populations and marine ecosystem dynamics: navigating the “predator pit” topographies. *Prog Oceanogr* 68:271–288. <https://doi.org/10.1016/j.pocean.2006.02.004>
- Bascompte J (2009) Disentangling the web of life. *Science* 325:416–419. <https://doi.org/10.1126/science.1170749>
- Bax N, Williamson A, Aguero M, Gonzalez E, Geeves W (2003) Marine invasive alien species: a threat to global biodiversity. *Mar Policy Emerg Issues Oceans Coasts Islands* 27:313–323. [https://doi.org/10.1016/S0308-597X\(03\)00041-1](https://doi.org/10.1016/S0308-597X(03)00041-1)
- Becker YA, Fioramonti NE, Dellabianca NA, Riccialdelli L (2021) Feeding ecology of the long finned pilot whale, *Globicephala melas edwardii*, in the southwestern Atlantic Ocean, determined by stable isotopes analysis. *Polar Biol* 44:1655–1667. <https://doi.org/10.1007/s00300-021-02908-2>
- Bornatowski H, Barreto R, Navia AF, de Amorim AF (2017) Topological redundancy and ‘small-world’ patterns in a food web in a subtropical ecosystem of Brazil. *Mar Ecol* 38:e12407. <https://doi.org/10.1111/maec.12407>
- Brickle P, Laptikhovsky V, Arkhipkin A, Portela J (2006) Reproductive biology of *Patagonotothen ramsayi* (Regan, 1913) (Pisces: Nototheniidae) around the Falkland Islands. *Polar Biol* 29:570–580. <https://doi.org/10.1007/s00300-005-0090-5>
- Büring T (2023) The ecological role and trophic relationships of seasonal cohorts of the Patagonian Squid (*Doryteuthis gahi*) in the marine ecosystem of the Falkland Islands. <https://doi.org/10.13140/RG.2.2.15667.81444>
- Büring T, Van Der Grient J, Pierce G, Bustamante P, Scotti M, Jones JB, Rocha F, Arkhipkin A (2024) Unveiling the wasp-waist structure of the Falkland shelf ecosystem: the role of *Doryteuthis gahi* as a keystone species and its trophic influences. *J Mar Biol Assoc UK* 104:e2. <https://doi.org/10.1017/S0025315423000887>
- Cardinale BJ, Duffy JE, Gonzalez A, Hooper DU, Perrings C, Venail P, Narwani A, Mace GM, Tilman D, Wardle DA, Kinzig AP, Daily GC, Loreau M, Grace JB, Larigauderie A, Srivastava DS, Naeem S (2012) Biodiversity loss and its impact on humanity. *Nature* 486:59–67. <https://doi.org/10.1038/nature11148>
- Csárdi G, Nepusz T, The igraph team (2023) Igraph enables fast and robust network analysis across programming languages. *arXiv*. <https://arxiv.org/abs/2311.10260>
- Céspedes R, Vargas C, Adasme L (2016) Monitoreo de la pesquería del bacalao de profundidad (*Dissostichus eleginoides*): observadores científicos y sus desafíos. *An Inst Patagon* 44:39–48. <https://doi.org/10.4067/S0718-686X2016000300004>
- Chadès I, Martin TG, Nicol S, Burgman MA, Possingham HP, Buckley YM (2011) General rules for managing and surveying networks of pests, diseases, and endangered species. *Proc Natl Acad Sci* 108:8323–8328. <https://doi.org/10.1073/pnas.1016846108>
- Cirtwill AR, Dalla Riva GV, Gaiarsa MP, Bimler MD, Cagua EF, Coux C, Dehling DM (2018) A review of species role concepts in food webs. *Food Webs* 16:e00093. <https://doi.org/10.1016/j.fooweb.2018.e00093>
- Cleveland WS, Grosse E, Shyu WM (1992) Local regression models. In: Statistical models in S. Wadsworth and Brooks/Cole, p 68
- Cohen JE, Briand F (1984) Trophic links of community food webs. *Proc Natl Acad Sci* 81:4105–4109. <https://doi.org/10.1073/pnas.81.13.4105>
- Cohen JE, Briand F, Newman CM (2012) Community food webs: data and theory. Springer, Cham
- Coll M, Navarro J, Olson RJ, Christensen V (2013) Assessing the trophic position and ecological role of squids in marine ecosystems by means of food-web models. *Deep Sea Res Part II Top Stud Oceanogr the Role of Squids in Pelagic Ecosystems* 95:21–36. <https://doi.org/10.1016/j.dsr2.2012.08.020>

- Collins MA, Shreeve RS, Fielding S, Thurston MH (2008) Distribution, growth, diet and foraging behaviour of the yellow-fin notothen *Patagonotothen guntheri* (Norman) on the Shag Rocks shelf (Southern Ocean). *J Fish Biol* 72:271–286. <https://doi.org/10.1111/j.1095-8649.2007.01711.x>
- Collins M, Brickle P, Brown J, Belchier M (2010) The Patagonian Toothfish. Biology, ecology and fishery. *Adv Mar Biol* 58:227–300. <https://doi.org/10.1016/B978-0-12-381015-1.00004-6>
- Connan M, Cherel Y, Mayzaud P (2007) Lipids from stomach oil of procellariiform seabirds document the importance of myctophid fish in the Southern Ocean. *Limnol Oceanogr* 52:2445–2455. <https://doi.org/10.4319/lo.2007.52.6.2445>
- Cordone G, Marina TI, Salinas V, Doyle SR, Saravia LA, Momo FR (2018) Effects of macroalgae loss in an Antarctic marine food web: applying extinction thresholds to food web studies. *PeerJ* 6:e5531. <https://doi.org/10.7717/peerj.5531>
- Coux C, Rader R, Bartomeus I, Tylanakis JM (2016) Linking species functional roles to their network roles. *Ecol Lett* 19:762–770. <https://doi.org/10.1111/ele.12612>
- Cury P (2000) Small pelagics in upwelling systems: patterns of interaction and structural changes in “wasp-waist” ecosystems. *ICES J Mar Sci* 57:603–618. <https://doi.org/10.1006/jmsc.2000.0712>
- Dahood A, de Mutsert K, Watters GM (2020) Evaluating Antarctic marine protected area scenarios using a dynamic food web model. *Biol Conserv* 251:108766. <https://doi.org/10.1016/j.biocon.2020.108766>
- Davic R (2003) Linking keystone species and functional groups: a new operational definition of the keystone species concept. *Conserv Ecol*. <https://doi.org/10.5751/ES-00502-0701r11>
- De Santana CN, Rozenfeld AF, Marquet PA, Duarte CM (2013) Topological properties of polar food webs. *Mar Ecol Prog Ser* 474:15–26. <https://doi.org/10.3354/meps10073>
- Dellabianca NA, Torres MA, Rey AR (2018) Occurrence of dusky dolphin *lagenorhynchus obscurus* in Tierra Del Fuego archipelago over five decades. *Polar Biol* 41:1895–1900. <https://doi.org/10.1007/s00300-018-2315-4>
- Dellabianca NA, Torres MA, Ordoñez C, Fioramonti N, Raya Rey A (2023) Marine protected areas in the southern south-west Atlantic: insights from marine top predator communities. *Aquat Conserv Mar Freshw Ecosyst* 33:472–487. <https://doi.org/10.1002/aqc.3935>
- Delmas E, Besson M, Brice MH, Burkle LA, Riva GVD, Fortin MJ, Gravel D, Guimarães PR, Hembry D, Newman E, Olesen JM, Pires MM, Yeakel JD, Poisot T (2018) Analyzing ecological networks of species interactions. *Biol Rev*. <https://doi.org/10.1101/112540>
- Diez MJ, Cabreira AG, Madirolas A, Lovrich GA (2016) Hydroacoustical evidence of the expansion of pelagic swarms of *Munida gregaria* (Decapoda, Munididae) in the Beagle Channel and the Argentine Patagonian Shelf, and its relationship with habitat features. *J Sea Res* 114:1–12. <https://doi.org/10.1016/j.seares.2016.04.004>
- Diez MJ, Cabreira AG, Madirolas A, De Nascimento JM, Scioscia G, Schiavini A, Lovrich GA (2018) Winter is cool: spatio-temporal patterns of the squat lobster *Munida gregaria* and the Fuegian sprat *Sprattus fuegensis* in a sub-Antarctic estuarine environment. *Polar Biol* 41:2591–2605. <https://doi.org/10.1007/s00300-018-2394-2>
- Diez M.J, Pérez-Barros P, Romero CM, Scioscia G, Tapella F, Cabreira AG, Madirolas A, Raya Rey A, Lovrich GA (2012) Pelagic swarms and beach strandings of the squat lobster *Munida gregaria* (Anomura: Munididae) in the Beagle Channel, Tierra del Fuego. *Polar Biol* 35:973–983. <https://doi.org/10.1007/s00300-011-1144-5>
- Dudley N (2008) Guidelines for applying protected area management categories. IUCN. <https://doi.org/10.2305/IUCN.CH.2008.PAPS.2.en>
- Dunne JA, Williams RJ, Martinez ND (2002a) Food-web structure and network theory: the role of connectance and size. *Proc Natl Acad Sci* 99:12917–12922. <https://doi.org/10.1073/pnas.192407699>
- Dunne JA, Williams RJ, Martinez ND (2002b) Network structure and biodiversity loss in food webs: robustness increases with connectance. *Ecol Lett* 5:558–567. <https://doi.org/10.1046/j.1461-0248.2002.00354.x>
- Earn DJ, Levin SA, Rohani P (2000) Coherence and conservation. *Science* 290:1360–1364. <https://doi.org/10.1126/science.290.5495.1360>
- Erdős P, Rényi A (1959) On random graphs. *Publicationes Mathematicae* 6:290–297. <https://doi.org/10.1007/BF03037638>
- Estrada E (2007) Food webs robustness to biodiversity loss: the roles of connectance, expansibility and degree distribution. *J Theor Biol* 244:296–307. <https://doi.org/10.1016/j.jtbi.2006.08.002>
- Estrada E, Bodin Ö (2008) Using network centrality measures to manage landscape connectivity. *Ecol Appl* 18:1810–1825. <https://doi.org/10.1890/07-1419.1>
- Falabella V (2017) Área Marina Protegida Namuncurá-Banco Burdwood. Contribuciones para la línea de base y el plan de manejo
- Ferro MD, Odriozola F, Torres C (2021) Áreas marinas protegidas en Argentina. Evolución jurídica y contexto internacional. *Obs Medioambient* 24:111–133. <https://doi.org/10.5209/obmd.79517>
- Fischer L, Covatti Ale M, Deli Antoni M, Díaz de Astarloa JM, Delpiani G (2022) Feeding ecology of the longtail southern cod, *Patagonotothen ramsayi* (Regan, 1913) (Notothenioidei) in the Marine Protected Area Namuncurá-Burdwood Bank, Argentina. *Polar Biol* 45:1483–1494. <https://doi.org/10.1007/s00300-022-03082-9>
- Flores H, Atkinson A, Kawaguchi S, Kraff BA, Milinevsky G, Nicol S, Reiss C, Tarling GA, Werner R, Bravo Rebolledo E, Cirelli V, Cuzin-Roudy J, Fielding S, Groeneveld JJ, Haraldsson M, Lombana A, Marschoff E, Meyer B, Pakhomov EA, Rombolá E, Schmidt K, Siegel V, Teschke M, Tonkes H, Toullec JY, Trathan PN, Tremblay N, Van de Putte AP, van Franeker JA & Werner T (2012) Impact of climate change on Antarctic krill. *Marine Ecol Progress Series*. 458:1–19
- Funes M, Saravia LA, Cordone G, Iribarne OO, Galván DE (2022) Network analysis suggests changes in food web stability produced by bottom trawl fishery in Patagonia. *Sci Rep* 12:10876. <https://doi.org/10.1038/s41598-022-14363-y>
- Gaichas SK, Francis RC (2008) Network models for ecosystem-based fishery analysis: a review of concepts and application to the Gulf of Alaska marine food web. *Canadian Journal of Fisheries and Aquatic Sciences* 65(9):1965–1982. <https://doi.org/10.1139/F08-104>
- Galleguillos R, Ferrada S, Canales-Aguirre C, Aedo G, San Martín M, Astete S (2006) Unidades poblacionales de bacalao de profundidad (Informe Final No. 41)
- Gasalla MA, Rodrigues AR, Postuma FA (2010) The trophic role of the squid *Loligo plei* as a keystone species in the South Brazil Bight ecosystem. *ICES J Mar Sci* 67:1413–1424. <https://doi.org/10.1093/icesjms/fsq106>
- Gillarranz LJ, Rayfield B, Liñán-Cembrano G, Bascompte J, Gonzalez A (2017) Effects of network modularity on the spread of perturbation impact in experimental metapopulations. *Science* 357:199–201. <https://doi.org/10.1126/science.aal4122>
- Gjøsaeter J, Kawaguchi K (1980) A review of the world resources of mesopelagic fish. Food & Agriculture Org
- Gorini FL, Di Marco E, Wöhler OC (2007) Estadística pesquera de peces demersales australes en el Atlántico Sudoccidental (período 1980–2002) (Informe Técnico)

- Guihou K, Piola AR, Palma ED, Chidichimo MP (2020) Dynamical connections between large marine ecosystems of austral South America based on numerical simulations. *Ocean Sci* 16:271–290. <https://doi.org/10.5194/os-16-271-2020>
- Halpern BS, Warner RR (2003) Matching marine reserve design to reserve objectives. *Proc R Soc B Biol Sci* 270:1871–1878. <https://doi.org/10.1098/rspb.2003.2405>
- Hunt GL, McKinnell S (2006) Interplay between top-down, bottom-up, and wasp-waist control in marine ecosystems. *Prog Oceanogr* 68:115–124. <https://doi.org/10.1016/j.pocean.2006.02.008>
- Jordán F (2009) Keystone species and food webs. *Philos Trans R Soc B Biol Sci* 364:1733–1741. <https://doi.org/10.1098/rstb.2008.0335>
- Jordán F, Scheuring I (2004) Network ecology: topological constraints on ecosystem dynamics. *Phys Life Rev* 1:139–172. <https://doi.org/10.1016/j.plrev.2004.08.001>
- Kones JK, Soetaert K, van Oevelen D, Owino JO (2009) Are network indices robust indicators of food web functioning? A Monte Carlo approach. *Ecol Model* 220:370–382. <https://doi.org/10.1016/j.ecolmodel.2008.10.012>
- Kortsch S, Primicerio R, Fossheim M, Dolgov AV, Aschan M (2015) Climate change alters the structure of arctic marine food webs due to poleward shifts of boreal generalists. *Proc R Soc B Biol Sci* 282:20151546. <https://doi.org/10.1098/rspb.2015.1546>
- Kortsch S, Primicerio R, Aschan M, Lind S, Dolgov AV, Planque B (2019) Food-web structure varies along environmental gradients in a high-latitude marine ecosystem. *Ecography* 42:295–308. <https://doi.org/10.1111/ecog.03443>
- Lovrich GA, Thiel M (2011) Ecology, physiology, feeding and trophic role of squat lobsters. *The biology of squat lobsters* 183:222
- La Mesa M, Riginella E, Melli V, Bartolini F, Mazzoldi C (2016) Biological traits of a sub-Antarctic nototheniid, *Patagonotothen ramsayi*, from the Burdwood Bank. *Polar Biol* 39:103–111. <https://doi.org/10.1007/s00300-015-1663-6>
- Laffoley D, Baxter J, Day, J, Wenzel L, Bueno P, Zischka K (2019) Marine protected areas, pp 549–569. <https://doi.org/10.1016/B978-0-12-805052-1.00027-9>
- Lai YR, Atkinson A, Kawaguchi S, Krafft BA, Milinevsky G, Nicol S, Trathan PN, Werner T (2012) Food webs: reconciling the structure and function of biodiversity. *Trends Ecol Evol* 27(12):689–697. <https://doi.org/10.1016/j.tree.2012.08.005>
- Lindeman RL (1942) The trophic-dynamic aspect of ecology. *Ecology* 23:399–417. <https://doi.org/10.2307/1930126>
- Link J (2002) Does food web theory work for marine ecosystems? *Mar Ecol Prog Ser* 230:1–9. <https://doi.org/10.3354/meps230001>
- López-López L, Genner MJ, Tarling GA, Saunders RA, O’Gorman EJ (2022) Ecological networks in the Scotia Sea: structural changes across latitude and depth. *Ecosystems* 25:457–470. <https://doi.org/10.1007/s10021-021-00665-1>
- Lubchenco J, Palumbi SR, Gaines SD, Andelman S (2003) Plugging a hole in the ocean: the emerging of marine reserves. *Ecol Appl* 13:3–7. [https://doi.org/10.1890/1051-0761\(2003\)013\[0003:PAHITO\]2.0.CO;2](https://doi.org/10.1890/1051-0761(2003)013[0003:PAHITO]2.0.CO;2)
- Maestro M, Pérez-Cayeiro ML, Chica-Ruiz JA, Reyes H (2019) Marine protected areas in the 21st century: current situation and trends. *Ocean Coast Manag* 171:28–36. <https://doi.org/10.1016/j.ocecoaman.2019.01.008>
- Marina TI, Saravia LA, Cordone G, Salinas V, Doyle SR, Momo FR (2018) Architecture of marine food webs: to be or not be a ‘small-world.’ *PLoS ONE* 13:e0198217. <https://doi.org/10.1371/journal.pone.0198217>
- Marina T, Schloss I, Lovrich G, Boy C, Bruno D, Capitanio F, Delpiani S, Díaz De Astarloa J, Frayse C, García Alonso V, Raya Rey A, Schejter L, Spinelli M, Tatián M, Urteaga D, Ricciardelli L (2024) The complex network of trophic interactions in a sub-Antarctic oceanic Marine Protected Area. *Mar Ecol Prog Ser*. <https://doi.org/10.3354/meps14600>
- Martín J, Alonso G, Dragani W, Meyerjürgens J, Giesecke R, Cucco A, Fenco H (2023) General circulation and tidal wave propagation along the Beagle Channel. *J Mar Syst* 240:103889. <https://doi.org/10.1016/j.jmarsys.2023.103889>
- Martín González AM, Dalsgaard B, Olesen JM (2010) Centrality measures and the importance of generalist species in pollination networks. *Ecol Complex* 7:36–43. <https://doi.org/10.1016/j.ecocom.2009.03.008>
- Martínez N (1992) Constant connectance in community food webs. *Am Nat* 139:1208–1218. <https://doi.org/10.1086/285382>
- Martínez PA, Wöhler OC, Troccoli GH, Marco EJD (2023) Efectos del establecimiento de las áreas marinas protegidas en la pesquería argentina de la merluza negra (*Dissostichus eleginoides*). *Mar Fish Sci MAFIS* 36:267–288. <https://doi.org/10.47193/mafis.3632023010906>
- May RM (1986) The search for patterns in the balance of nature: advances and retreats. *Ecology* 67:1115–1126. <https://doi.org/10.2307/1938668>
- McCann K, Hastings A (1997) Re-evaluating the omnivory–stability relationship in food webs. *Proceedings of the Royal Society of London. Series B: Biological Sciences* 264(1385):1249–1254. <https://doi.org/10.1098/rspb.1997.0172>
- McDonald-Madden E, Sabbadin R, Game ET, Baxter PWJ, Chadès I, Possingham HP (2016) Using food-web theory to conserve ecosystems. *Nat Commun* 7:10245. <https://doi.org/10.1038/ncomms10245>
- Milano VN, Grandi MF, Schiavini ACM, Crespo EA (2020a) Recovery of South American fur seals from Fuegian Archipelago (Argentina). *Mar Mammal Sci* 36:1022–1032. <https://doi.org/10.1111/mms.12686>
- Milano VN, Grandi MF, Schiavini ACM, Crespo EA (2020b) Sea lions (*Otaria flavescens*) from the end of the world: insights of a recovery. *Polar Biol* 43:695–706. <https://doi.org/10.1007/s00300-020-02672-9>
- Montoya JM, Solé RV (2002) Small world patterns in food webs. *J Theor Biol* 214:405–412. <https://doi.org/10.1006/jtbi.2001.2460>
- Montoya JM, Pimm SL, Solé RV (2006) Ecological networks and their fragility. *Nature* 442:259–264. <https://doi.org/10.1038/nature04927>
- Mouquet N, Gravel D, Massol F, Calcagno V (2013) Extending the concept of keystone species to communities and ecosystems. *Ecol Lett* 16:1–8. <https://doi.org/10.1111/ele.12014>
- Murphy EJ, Watkins JL, Trathan PN, Reid K, Meredith MP, Thorpe SE, Johnston NM, Clarke A, Tarling GA, Collins MA, Forcada J, Shreeve RS, Atkinson A, Korb R, Whitehouse MJ, Ward P, Rodhouse PG, Enderlein P, Hirst AG, Martin AR, Hill SL, Staniland IJ, Pond DW, Briggs DR, Cunningham NJ, Fleming AH (2006) Spatial and temporal operation of the Scotia Sea ecosystem: a review of large-scale links in a krill centred food web. *Philos Trans R Soc B Biol Sci* 362:113–148. <https://doi.org/10.1098/rstb.2006.1957>
- Navarro J, Coll M, Somes CJ, Olson RJ (2013) Trophic niche of squids: insights from isotopic data in marine systems worldwide. *Deep Sea Res Part II Top Stud Oceanogr* 95:93–102. <https://doi.org/10.1016/j.dsr2.2013.01.031>
- Navia AF, Cruz-Escalona VH, Giraldo A, Barausse A (2016) The structure of a marine tropical food web and its implications for ecosystem-based fisheries management. *Ecol Model* 328:23–33. <https://doi.org/10.1016/j.ecolmodel.2016.02.009>
- Ordoñez C, Diez MJ, Torres MA, Dellabianca NA (2020) Uso de hábitat del delfín austral (*Lagenorhynchus australis*) en el Canal Beagle, Tierra del Fuego, Argentina. *Mastozool Neotropical* 27:247–252. <https://doi.org/10.31687/saremMN.20.27.2.0.16>
- Padovani LN, Viñas MD, Sánchez F, Mianzan H (2012) Amphipod-supported food web: *Themisto gaudichaudii*, a key food resource

- for fishes in the southern Patagonian Shelf. *J Sea Res* 67:85–90. <https://doi.org/10.1016/j.seares.2011.10.007>
- Palma ED, Matano RP, Piola AR (2004) A numerical study of the Southwestern Atlantic Shelf circulation: barotropic response to tidal and wind forcing. *J Geophys Res Oceans*. <https://doi.org/10.1029/2004JC002315>
- Palma FI, Bozzano G, Principi S, Isola JJ, Ormazabal JP, Esteban FD, Tassone AA (2021) Geomorphology and sedimentary processes on the Sloggett Canyon, Northwestern Scotia Sea, Argentina. *J South Am Earth Sci* 107:103136. <https://doi.org/10.1016/j.jsames.2020.103136>
- Pascual M, Dunne JA (eds) (2006) Ecological networks: linking structure to dynamics in food webs. Oxford University Press, Oxford
- Pauly D, Palomares ML (2005) Fishing down marine food web: it is far more pervasive than we thought. *Bull Mar Sci* 76
- Pauly D, Christensen V, Gu  nette S, Pitcher TJ, Sumaila UR, Walters CJ, Watson R, Zeller D (2002) Towards sustainability in world fisheries. *Nature* 418:689–695. <https://doi.org/10.1038/nature01017>
- Peterson RG, Whitworth T (1989) The subantarctic and polar fronts in relation to deep water masses through the southwestern Atlantic. *J Geophys Res Oceans* 94:10817–10838. <https://doi.org/10.1029/JC094iC08p10817>
- Pimm, SL (1982) The length of food chains. In: Food webs Population and community biology. Springer, Dordrecht. https://doi.org/10.1007/978-94-009-5925-5_6
- Piola AR, Gordon AL (1989) Intermediate waters in the southwest South Atlantic. *Deep Sea Res Part Oceanogr Res Pap* 36:1–16. [https://doi.org/10.1016/0198-0149\(89\)90015-0](https://doi.org/10.1016/0198-0149(89)90015-0)
- Presta ML, Riccialdelli L, Bruno DO, Castro LR, Fioramonti NE, Florent  n OV, Berghoff CF, Capitanio FL, Lovrich GA (2023) Mesozooplankton community structure and trophic relationships in an austral high-latitude ecosystem (Beagle Channel): the role of bottom-up and top-down forces during springtime. *J Mar Syst* 240:103881. <https://doi.org/10.1016/j.jmarsys.2023.103881>
- Radovich VS (2019) Gesti  n ambiental del Mar en la Rep  blica Argentina: reflexiones sobre los parques interjurisdiccionales marino costeros y las   reas marinas protegidas. *Gest Ambiente* 22:67–78. <https://doi.org/10.15446/ga.v22n1.77181>
- Riccialdelli L, Becker Y, Fioramonti N, Torres M, Bruno D, Raya Rey A, Fern  ndez D (2020) Trophic structure of southern marine ecosystems: a comparative isotopic analysis from the Beagle Channel to the oceanic Burdwood Bank area under a wasp-waist assumption. *Mar Ecol Prog Ser* 655:1–27. <https://doi.org/10.3354/meps13524>
- Rodr  guez ID, Marina TI, Schloss IR, Saravia LA (2022) Marine food webs are more complex but less stable in sub-Antarctic (Beagle Channel, Argentina) than in Antarctic (Potter Cove, Antarctic Peninsula) regions. *Mar Environ Res* 174:105561. <https://doi.org/10.1016/j.marenvres.2022.105561>
- Romero SI, Piola AR, Charo M, Garc  a CAE (2006) Chlorophyll- *a* variability off Patagonia based on SeaWiFS data. *J Geophys Res Oceans*. <https://doi.org/10.1029/2005JC003244>
- Rodr  guez ID, Saravia LA (2024) Potter Cove's Heavyweights: Estimation of Species' Interaction Strength of an Antarctic Food Web. *Ecol Evol* 14(11):e70389 <https://doi.org/10.1002/ece3.70389>
- Saba GK, Fraser WR, Saba VS, Iannuzzi RA, Coleman KE, Doney SC, Ducklow HW, Martinson DG, Miles TN, Patterson-Fraser DL, Stammerjohn SE, Steinberg DK, Schofield OM (2014) Winter and spring controls on the summer food web of the coastal West Antarctic Peninsula. *Nat Commun* 5:4318. <https://doi.org/10.1038/ncomms5318>
- Sabatini M, Reta R, Matano R (2004) Circulation and zooplankton biomass distribution over the southern Patagonian shelf during late summer. *Cont Shelf Res* 24:1359–1373. <https://doi.org/10.1016/j.csr.2004.03.014>
- Sabatini ME, Alvarez Colombo GL (2007) Seasonal pattern of zooplankton biomass in the Argentinian shelf off Southern Patagonia (45  –55   S). <https://doi.org/10.3989/scimar.2001.65n121>
- Sabatini ME, Akselman R, Reta R, Negri RM, Lutz VA, Silva RI, Segura V, Gil MN, Santinelli NH, Sastre AV, Daponte MC, Anta  li JC (2012) Spring plankton communities in the southern Patagonian shelf: hydrography, mesozooplankton patterns and trophic relationships. *J Mar Syst* 94:33–51. <https://doi.org/10.1016/j.jmarsys.2011.10.007>
- S  nchez RP, Remeslo A, Madirolas A, de Ciechowski JD (1995) Distribution and abundance of post-larvae and juveniles of the Patagonian sprat, *Sprattus fuegensis*, and related hydrographic conditions. *Fish Res* 23:47–81. [https://doi.org/10.1016/0165-7836\(94\)00339-X](https://doi.org/10.1016/0165-7836(94)00339-X)
- Saporiti F, Bearhop S, Vales D, Silva L, Zenteno L, Tavares M, Crespo E, Cardona L (2015) Latitudinal changes in the structure of marine food webs in the Southwestern Atlantic Ocean. *Mar Ecol Prog Ser* 538:23–34. <https://doi.org/10.3354/meps11464>
- Saravia LA (2023) Multiweb: Ecological network analyses including multiplex networks. <https://github.com/saravia/multiweb>. Accessed Nov 1 2024
- Saunders RA, Hill SL, Tarling A, Murphy EJ (2019) Myctophid fish (Family *Myctophidae*) are central consumers in the Food Web of the Scotia Sea (Southern Ocean). *Front Mar Sci*. <https://doi.org/10.3389/fmars.2019.00530>
- Saunders RA, Collins MA, Ward P, Stowasser G, Hill SL, Shreeve R, Tarling GA (2015) Predatory impact of the myctophid fish community on zooplankton in the Scotia Sea (Southern Ocean). *Mar Ecol Prog Ser* 541:45–64. <https://doi.org/10.3354/meps11527>
- Shreeve RS, Collins MA, Tarling GA, Main CE, Ward P, Johnston NM (2009) Feeding ecology of myctophid fishes in the northern Scotia Sea. *Mar Ecol Prog Ser* 386:221–236. <https://doi.org/10.3354/meps08064>
- Sih A, Jonsson BG, Luikart G (2000) Habitat loss: ecological, evolutionary and genetic consequences. *Trends Ecol Evol* 15:132–134. [https://doi.org/10.1016/S0169-5347\(99\)01799-1](https://doi.org/10.1016/S0169-5347(99)01799-1)
- Stouffer DB, Bascompte J (2011) Compartmentalization increases food-web persistence. *Proc Natl Acad Sci* 108:3648–3652. <https://doi.org/10.1073/pnas.1014353108>
- Th  bault E, Fontaine C (2010) Stability of Ecological communities and the architecture of mutualistic and trophic networks. *Science* 329:853–856. <https://doi.org/10.1126/science.1188321>
- Thompson RM, Hemberg M, Starzomski BM, Shurin JB (2007) Trophic levels and trophic tangles: the prevalence of omnivory in real food webs. *Ecology* 88:612–617. <https://doi.org/10.1890/05-1454>
- Thompson RM, Brose U, Dunne JA, Hall RO, Hladysz S, Kitching RL, Martinez ND, Rantala H, Romanuk TN, Stouffer DB, Tylianakis JM (2012) Food webs: reconciling the structure and function of biodiversity. *Trends Ecol Evol* 27:689–697. <https://doi.org/10.1016/j.tree.2012.08.005>
- Tombesi ML, Rabuffetti F, Lovrich GA (2020) Las   reas marinas protegidas en la Argentina. *Lupa Colecc. Fueguina Divulg Cient  fica* 16:2–7. <https://www.coleccionlalupa.com.ar/index.php/lalupa/article/view/79>
- Troc  oli GH, Aguilar E, Mart  nez PA, Belleggia M (2020) The diet of the Patagonian toothfish *Dissostichus eleginoides*, a deep-sea top predator off Southwest Atlantic Ocean. *Polar Biol* 43:1595–1604. <https://doi.org/10.1007/s00300-020-02730-2>
- Tundi Agardy M (1994) Advances in marine conservation: the role of marine protected areas. *Trends Ecol Evol* 9:267–270. [https://doi.org/10.1016/0169-5347\(94\)90297-6](https://doi.org/10.1016/0169-5347(94)90297-6)
- United Nations (2018) The 2030 Agenda and the Sustainable Development Goals: an opportunity for Latin America and the Caribbean. United Nations, Santiago de Chile

- United Nations (1992) Convention on biological diversity. https://treaties.un.org/pages/ViewDetails.aspx?chapter=27&mtdsg_no=XXVII-8&src=TREATY
- Van der Grient J, Morley S, Arkhipkin A, Bates J, Baylis A, Brewin P, Harte M, White JW, Brickley P (2023) The Falkland Islands marine ecosystem: A review of the seasonal dynamics and trophic interactions across the food web Elsevier 1–68
- Voigt I, Henrich R, Preu BM, Piola AR, Hanebuth TJJ, Schwenk T, Chiessi CM (2013) A submarine canyon as a climate archive—interaction of the Antarctic Intermediate Water with the Mar del Plata Canyon (Southwest Atlantic). *Mar Geol* 341:46–57. <https://doi.org/10.1016/j.margeo.2013.05.002>
- Watts DJ, Strogatz SH (1998) Collective dynamics of ‘small-world’ networks 393
- Watters GM, Hinke JT, Reiss CS (2020) Long-term observations from Antarctica demonstrate that mismatched scales of fisheries management and predator-prey interaction lead to erroneous conclusions about precaution. *Abstract Sci Rep* 10(1):2314. <https://doi.org/10.1038/s41598-020-59223-9>
- Wöhler OC, Cordo HD, Hansen JE (2001) Rendimiento Por Unidad de Esfuerzo de Pesqueros Comerciales
- Yates P, Ziegler P, Welsford D, McIvor J, Farmer B, Woodcock E (2018) Spatio-temporal dynamics in maturation and spawning of Patagonian toothfish *Dissostichus eleginoides* on the sub-Antarctic Kerguelen Plateau. *J Fish Biol* 92:34–54. <https://doi.org/10.1111/jfb.13479>
- Young JW, Hunt BPV, Cook TR, Llopiz JK, Hazen EL, Pethybridge HR, Ceccarelli D, Lorrain A, Olson RJ, Allain V, Menkes C, Patterson T, Nicol S, Lehodey P, Kloser RJ, Arrizabalaga H, Anela Choy C (2015) The trophodynamics of marine top predators: current knowledge, recent advances and challenges. *Deep Sea Res Part II Top Stud Oceanogr Impacts of Clim Mar Top Predat* 113:170–187. <https://doi.org/10.1016/j.dsr2.2014.05.015>
- Zhu J, Zhu G (2022) Trophic linkage between mackerel icefish (*Champsocephalus gunnari*) and Antarctic krill (*Euphausia superba*) at South Georgia. *Fish Res* 253:106366. <https://doi.org/10.1016/j.fishres.2022.106366>

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